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American Journal of Hematology

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Facts and Misfacts on D-Dimer Testing. Consensus Guidance From the Italian Society on Thrombosis and Hemostasis (SISET)

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Received: 17 May 2025 | **Revised:** 4 September 2025 | **Accepted:** 13 September 2025

Funding: The authors received no specific funding for this work.

Keywords: acute aortic syndrome | deep-vein thrombosis | direct oral anticoagulants | optimal duration of secondary thromboprophylaxis | pulmonary embolism | vitamin K antagonists

ABSTRACT

D-dimer defines degradation products derived from the proteolysis mediated by plasmin on cross-linked fibrin. The evidence-based use of D-dimer in some conditions has been consolidated. Currently, however, there is an entrenched prescription of D-dimer testing to screen otherwise healthy subjects that may induce prescribing physicians to start time-consuming diagnostic procedures that may generate patient anxiety/distress and often fail to reach meaningful conclusions for patients' management. To make things worse, the incidental finding of unexplained elevated D-dimer is often, and inadvertently, taken as an index of venous thromboembolism (VTE) that may induce unnecessary initiation of thromboprophylaxis. Finally, population studies showed that elevated D-dimer is variably associated with arterial thrombosis. However, the above studies do not allow establishing a direct causal relationship between elevated D-dimer and disease, which makes D-dimer measurement hardly applicable in individual patients. With this as background, the Italian Society on Thrombosis and Hemostasis appointed a panel to prepare a guidance on the use of D-dimer based on the literature and expert opinion. The main conclusions of the panel can be summarized as follows: (i) D-dimer can be reliably used in combination with pre-test clinical probability to exclude VTE in symptomatic outpatients. (ii) It may help establish the optimal duration of secondary VTE prevention with vitamin K antagonists, although the panel felt that this area needs further investigation when dealing with direct oral anticoagulants. (iv) The use of D-dimer in combination with the pre-test probability score can be used to safely rule out acute aortic syndromes in the emergency setting without further imaging tests.

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1 | Introduction

D-dimer defines a group of degradation products derived from the proteolysis mediated by plasmin on fibrin cross-linked by factor XIIIa (Figure 1). Because of the above characteristics, D-dimer is a reliable biomarker of fibrin deposition and thrombus formation. In the past, D-dimer was measured by immunoassays employing polyclonal antibodies that reacted not only with D-dimer, but also with intact fibrinogen and other fragments derived from digested fibrinogen. More recently, monoclonal antibodies (MoAbs) directed to new epitopes exposed on D-dimer were developed and largely employed to construct immunochemistry assays with increased specificity for D-dimer. Presently, there are many methods to measure D-dimer, all exploiting the same analytical principle, but in different formats. The enzyme-linked immunosorbent assay (ELISA) shows good sensitivity and specificity, but it is time-consuming and needs special equipment and expertise. Presently, the most widely used assays are based on latex particles, coated with MoAbs to D-dimer [1]. In the absence of D-dimer, particles remain in suspension. In the presence of D-dimer, particles agglutinate and the suspension clears up. The reaction is monitored by the photo-optical system employed by ordinary coagulometers that translates changes of optical density into D-dimer concentrations by comparison with a certified standard. Methods based on latex particles are rapid and their sensitivity/specificity is comparable to those of the ELISA. Finally, D-dimer assays based on detecting agglutination of latex particles or red blood cells by visual inspection were also developed and are commercially available. These methods are rapid, but poorly sensitive when compared to ELISA or automated latex particle methods. Their results are also operator dependent.

When interpreting results for D-dimer, it is important to realize that there is still a lack of methodological standardization [2]. This is due to several reasons. First, the variety of D-dimer molecular species that are produced by the degradation of stabilized fibrin react differently with MoAbs used to prepare commercial assays. Second, there are no established international standards for D-dimer to prepare calibration curves capable of harmonizing results across methods. Third, there is no standardized expression of results; some methods report results as mass concentrations related to fibrinogen-equivalent units and others as D-dimer units [3]. All the above issues make D-dimer results produced by different methods hardly comparable, especially when concentrations are relatively high. It is therefore imperative to refer to decision levels specific for the method being used. In this respect, it is important to realize that there are two decision levels that can be used to interpret results of D-dimer. The reference value that can be calculated by measuring D-dimer for healthy subjects typically corresponds to the mean + 2SD (or the 95th centile) of the results distribution. In contrast, the cut-off value is derived from the investigation of patients suspected of having VTE, including deep vein thrombosis (DVT) and pulmonary embolism (PE), who are investigated with imaging to rule in or out VTE. Typically, cut-off values corresponding to the D-dimer level that yields a sensitivity close to 100% and reasonable specificity for VTE are chosen [4]. The cut-off value should be used to exclude VTE. The reference value should be used to interpret results when investigating patients suspected of having disseminated intravascular coagulation (DIC) or other conditions not directly related to thrombosis. It is important to realize that reference and cut-off values may or may not coincide for the same method.

D-dimer, being highly sensitive, may be elevated in many clinical conditions in addition to VTE (Figure 2). However, the specificity is relatively poor and therefore it is a misconception to believe that elevated D-dimer is invariably associated with VTE.

There are clinical conditions where the value of D-dimer in clinical practice has been substantiated by evidence, but in others, the value of D-dimer has been plausibly claimed although not yet confirmed by clinical observations. For example, there is currently an entrenched unjustified prescription of D-dimer testing to screen otherwise healthy subjects that may induce prescribing physicians to start time-consuming diagnostic procedures that may generate patient anxiety/distress and often fail to reach meaningful conclusions for patients' management. To make things worse, the incidental finding of unexplained elevated D-dimer is often and inadvertently taken

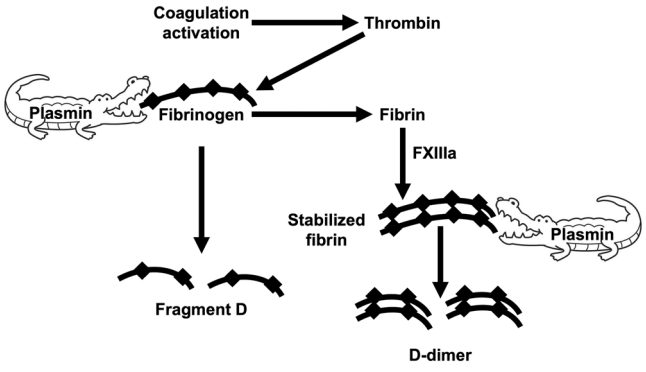


FIGURE 1 | Schematic representation of the fibrin degradation products mediated by plasmin. FXIIIa, activated factor XIII.

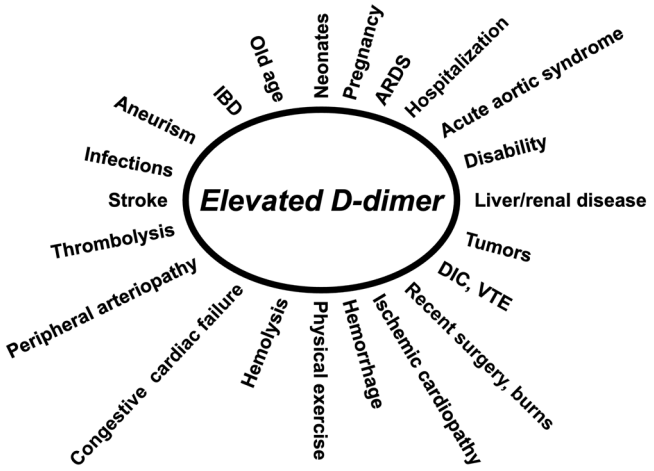


FIGURE 2 | Clinical conditions associated with increased levels of D-dimer. ARDS, acute respiratory distress syndrome; DIC, disseminated intravascular coagulation; VTE, venous thromboembolism.

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as an index of VTE that may induce unnecessary initiation of thromboprophylaxis. Finally, population studies showed that elevated D-dimer is variably associated with arterial thrombosis. However, the above studies do not allow establishing a direct causal relationship between elevated D-dimer and disease, which makes D-dimer measurement hardly applicable in individual patients. With this as background, the Italian Society on Thrombosis and Hemostasis (SISET) appointed a panel of experts (A Tripodi coordinator and M Marchetti methodologist) to prepare a guidance on the use of D-dimer testing based on the literature and expert opinion. It should be emphasized that this is not a guideline document, as it does not report a systematic review of the literature and meta-analyses with the usual forest plots, but a consensus guidance based on expert opinion and pertinent clinical studies.

The first part of the guidance aims to focus on the clinical conditions for which there is evidence for the value of D-dimer (e.g., exclusion of VTE, to establish the optimal duration of secondary VTE thromboprophylaxis, and diagnosis of acute aortic syndrome).

The second part discusses other conditions where the measurement of D-dimer is considered plausibly useful, but there is not yet evidence from clinical observations.

2 | Methods and Search for Evidence

For both parts of the guidance, we followed the same method. EMBASE and PubMed were searched to scan high-quality evidence supporting the value of D-dimer. Based on evidence, we formulated statements that were later anonymously voted on by the panelists.

For those situations where the clinical value of D-dimer was advocated, but not yet substantiated by clinical observations, we elaborated a set of seven questions that may represent possible scenarios encountered in clinical practice.

The project strived to adhere to the standard guideline method GRADE; therefore, clinical questions were framed according to the Population-Intervention-Comparator-Outcome (PICO) format (Table 1).

The target POPULATION is adult outpatients asymptomatic for thrombosis, who have unexplained elevation of D-Dimer above the cut-off value used for the diagnostic work-up of VTE, after exclusion of possible contributions to increased D-Dimer from heterophile antibodies (i.e., weak antibodies with multi-specific activities against poorly defined antigens) or antibodies against rheumatoid factor or problems related to poor blood drawing. Target patients had also to be free from one or more of the conditions reported in Table 2 in which D-Dimer may be elevated because of unknown reasons and for which giving guidance on testing is difficult. The typical example is DIC, which is currently searched by including D-Dimer among the test panel, even though the official ISTH guidelines on DIC [5] prescribe testing for fibrin/fibrinogen degradation products and not D-Dimer.

TABLE 1 | Characteristics of PICO questions.

Population	Intervention	Outcomes
First incidental finding of unexplained elevation of D-dimer levels	Compression Ultrasonography (CUS), CT-scan	Unsuspected VTE: detection & treatment
	Repeat D-dimer testing after 2–3 months	Unsuspected VTE: Detection & treatment
	Limited and extended cancer screening[a]	Occult cancer: Detection & treatment
	Testing for inherited or acquired thrombophilia[b]	Unsuspected VTE: detection & treatment
	Testing for laboratory risk factors of arterial thrombosis[c]	Cardiovascular events[d] Detection and prevention
	Antithrombotic pharmacological (LMWH) prophylaxis of VTE	VTE prevention
	Instrumental test[e] for atherosclerosis in patients with moderate-to-high 10-years cardiovascular risk according to validated risk assessment models	Cardiovascular events[d]: Detection & prevention

[a]Medical history and physical examination, laboratory and imaging investigations (e.g., complete blood count, calcium, urinalysis, and liver function tests, chest x-ray, abdomen US). Laboratory investigations plus chest and abdomen/pelvis CT scan.
[b]Protein C, protein S, antithrombin, Factor V Leiden, and Prothrombin 20210A mutations, Lupus anticoagulant, anticardiolipin IgG and IgM antibodies, antiβ-2GPI IgG and IgM antibodies.
[c]Fibrinogen, FVII, homocysteine, high-sensitivity C-reactive protein, lupus anticoagulant, anticardiolipin, and antiβ-2GPI antibodies.
[d]Ischemic stroke, acute myocardial infarction, acute coronary syndrome, symptomatic arterial peripheral disease.
[e]Arterial ultrasonography (carotid or lower limb), ECG, echocardiography.

The addressed INTERVENTIONS span from tests to antithrombotic prophylaxis.

DIAGNOSTIC INTERVENTIONS include imaging tests aimed at ruling out VTE, cancer screening procedures, and laboratory tests aimed at assessing risk factors for venous or arterial thromboses.

The major PICO question OUTCOMES include detection and treatment of asymptomatic venous or arterial thromboses as well as detection and treatment of occult cancer.

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STATEMENTS

- Statements were formulated for each PICO question based on number needed to test and uncertainty in the available direct and indirect evidence.
- Statements underwent anonymous approval process according to Delphi consensus method and were judged as "*approved*" if 75% or more of the panelists agreed (*either % strongly agreed or % strongly agreed + agreed*) in their current content.
- Interventions were "*strongly suggested against*" if the expected clinical benefit was not clinically relevant or uncertainty was disproportionate to the expected clinical benefit.
- Interventions were "*strongly suggested*" if the expected clinical benefit was clinically relevant and uncertainty was proportionate to the expected clinical benefit.
- Interventions were "*suggested*" if the expected clinical benefit was clinically relevant but uncertainty was disproportionate to the expected clinical benefit.

3 | Evidence-Based Use of D-Dimer in Clinical Practice

There are clinical conditions where the value of D-dimer in clinical practice was substantiated by evidence (e.g., exclusion of VTE, to establish the optimal duration of secondary VTE thromboprophylaxis and diagnosis of acute aortic syndrome).

3.1 | D-Dimer to Exclude VTE

VTE is a relatively frequent disease affecting nearly 10 million people every year worldwide [6]. The diagnosis based on clinical grounds is unreliable, and imaging is needed, which is, however, time-consuming and expensive. Over the last three decades, diagnostic strategies incorporating pre-test clinical probability, D-dimer measurement, and imaging were developed and validated. The successful story of D-dimer in the diagnosis of VTE began in 1991 when Bounameux et al. [4] investigated patients presenting with symptoms of PE. These patients underwent imaging to rule in or out PE and were concomitantly subjected to D-dimer measurement. Upon stratification of positive/negative PE vs. D-dimer concentrations, it appeared that the median D-dimer level was higher in patients with than without PE. There was, however, a relatively large results overlap, whereby patients with PE could not be distinguished with certainty from those without. Interestingly, upon establishing an arbitrary cut-off D-dimer value corresponding to 500 μg/L, the authors observed that only one patient with PE had D-dimer concentration below cut-off. This observation, which was also validated for DVT, paved the way to the concept that, although D-dimer cannot be used to confirm VTE, it can be used to exclude VTE. Other evidence came later from the literature (reviewed in ref [7]) However,

TABLE 2 | Target population considered in this guidance.

Target population
Adult outpatients asymptomatic for thrombosis who have unexplained elevation of D-dimer above the cut-off value used for the diagnostic work-up of VTE[a] and none of the following conditions
No known active cancer
Hemangioma or varices
Ongoing or recent pregnancy
Recent (< 6 weeks) delivery or abortion
Treatment leading to ovarian hyperstimulation
Recent (< 3 months) surgery
Overt major or clinically relevant bleeding
Trauma
Acute or chronic infection/inflammatory diseases
Covid-19 infection
Vaccination
Inherited dysfibrinogenemia
Disseminated intravascular coagulation

[a]After exclusion of possible contributions to increased D-dimer from heterophile antibodies (i.e., weak antibodies with multi-specific activities against poorly defined antigens) or antibodies against rheumatoid factor or problems related to poor blood drawing.

D-dimer cannot be used as a standalone parameter to exclude VTE, as the risk of incurring in false-positive or false-negative VTE is relatively high (Table 3). Elevated D-dimer and hence false-positive VTE can, in fact, be due to conditions other than intravascular fibrin deposition (Figure 2). On the other hand, false-negative VTE based on low D-dimer can occur because of poor test sensitivity, inaccurate cut-off values, hypofibrinolysis, ongoing antithrombotic treatment, or symptoms older than 7–10 days (Table 3).

3.1.1 | Diagnostic Strategies to Exclude VTE

Among the strategies that were developed to limit false-positive and false-negative VTE, the most convenient is the measurement of D-dimer combined with the pre-test clinical VTE probability and further investigation with imaging [8–10]. The pre-test clinical VTE probability is assessed by a score (e.g., Wells score) incorporating clinical signs, symptoms, and risk factors for DVT or PE. This strategy can be applied to symptomatic outpatients. Based on a relatively simple algorithm, if D-dimer is below cut-off (typically 500 or 250 μg/L, depending on the method) and the pre-test VTE clinical probability is low, VTE is unlikely and can be excluded with an accuracy close to 100%. On the other hand, if the clinical probability is relatively high, patients must be subjected to imaging, regardless of the D-dimer levels. The above strategy was validated by numbers of management studies showing that very few

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TABLE 3 | Causes of false-positive or false-negative venous thromboembolism (VTE).

Causes of false-positive VTE with elevated D-dimer
Recent surgery
Hemorrhage
Trauma
Malignancy
Sepsis
Age
Pregnancy
Disseminated intravascular coagulation
Atrial fibrillation
Hematomas and ascites
Hospitalization
Physical exercise
Varices
Hemangiomas
Autoimmune diseases
Inflammation
Acute aortic dissection
Cirrhosis
Diabetes
Renal diseases
Inherited dysfibrinogenemia
Causes of false-negative VTE with normal D-dimer
Poor test sensitivity
Inaccurate cut-off
Hypofibrinolysis
Symptoms older than 7–10 days
Antithrombotic therapy

patients developed recurrent VTE after exclusion following the application of the algorithm [11–15].

3.1.2 | Conditions Where Exclusion of VTE Is Complicated

It is important to realize that there are conditions in which the above strategy was not conclusively validated.

Exclusion of VTE during pregnancy is still a matter of investigation. D-dimer increases along with the pregnancy, and cut-off values valid for this condition are not established [16]. Furthermore, no specific pre-test probability scores have been developed. A recent metanalysis [17] of two studies [18, 19], included 893 pregnant women suspected of having PE. The results showed that the application of the Wells score with fixed and adapted D-dimer threshold, as well as the YEARS algorithm, could rule out acute PE. However, the metanalysis had some limitations. The small sample size could have resulted in relatively large confidence intervals of the estimates. The two individual studies applied different diagnostic strategies (e.g., Wells score or YEARS algorithm); hence, a proportion of the patients included in the metanalysis were not managed by the Wells score or by the YEARS algorithm and were reclassified post hoc. Although compression ultrasonography was included in the diagnostic algorithms of the two individual studies, it was used at a different stage and with a different indication. Finally, while the proportion of inconclusive test results was a predefined endpoint in one study, in the other, these were only defined as such when a treatment decision could not be made based on the quality of the scan.

VTE exclusion by means of D-dimer and Wells score is not firmly validated in hospitalized patients [20]. Most of these patients have elevated levels of D-dimer without documented VTE.

Finally, the accuracy of the VTE exclusion by means of D-dimer and Wells score is limited also in patients with cancer. In most of these patients, D-dimer is increased by the underlying disease and is poorly specific for VTE. Hence, it is likely that a considerable proportion of cancer patients will undergo imaging tests [21].

3.1.3 | Age-Adjusted Cut-Off Values

Over the years, it was observed that the diagnostic accuracy of the combination of D-dimer with the pre-test clinical probability is not effective in old patients [22]. D-dimer is, in fact, increasing with age and therefore also the cut-off value for VTE increases. For example, application of the usual cut-off of 500 μg/L in an 80 years-old patient would probably lead to the erroneous conclusion of positive VTE. Studies aimed at increasing the diagnostic efficacy of D-dimer in old patients were carried out, and the simplest strategy that emerged is the determination of individual age-adjusted cut-off based on the patient's age multiplied by 10 μg/L in patients > 50 years [22]. Accordingly, a 70 years old patient will have a cut-off of 700 μg/L instead of the regular cut-off of 500 μg/L. A collaborative study including 2818 consecutive outpatients with DVT symptoms confirmed that the age-adjusted cut-off of D-dimer combined with pre-test clinical probability increases the percentage of older patients in whom DVT can be safely excluded [22].

3.1.4 | Choice of Methods

Since there are many commercial methods for D-dimer, the choice should be based on the following characteristics [13].

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(i) Quantitative methods should be preferred over qualitative ones. (ii) The method of choice should have a sensitivity for VTE possibly close to 100% and a reasonable specificity. (iii) Cut-off values should be determined according to the Food and Drug Administration (FDA) recommendations by investigating patients who presented symptoms of VTE ruled in or out by imaging. Validated methods are certified by the FDA.

STATEMENTS

- We *strongly suggest the use of D-dimer in combination with pre-test clinical probability* (*score of Wells*) to safely exclude VTE in symptomatic outpatients (*Anonymous panelist vote: 83.3% strongly agreed*).
- If the pre-test clinical probability for VTE is low and D-dimer is below cut-off, VTE can be safely excluded (*83.3% strongly agreed*).
- If the pre-test clinical probability for VTE is high, imaging must be performed to confirm VTE, regardless of the D-dimer levels (*66.7% strongly agreed + 25% agreed = 91.7%*).
- We *strongly suggest the use of age-adjusted cut-off D-dimer for the exclusion of VTE* (*66.7% strongly agreed + 33.3% agreed = 100%*).
- We *suggest the use of quantitative rather than qualitative methods*. Sensitivity for VTE should be close to 100% with reasonable specificity (*75% strongly agreed*).
- We *strongly suggest* methods certified by FDA (*75% strongly agreed*).

3.2 | D-Dimer to Establish the Optimal Duration of Secondary Thromboprophylaxis for VTE

VTE tends to recur [23] and therefore patients after the first VTE event are given thromboprophylaxis with low molecular weight heparin (LMWH), vitamin K antagonists (VKAs), or direct oral anticoagulants (DOACs). Depending on the risk of recurrence and hemorrhage in individual patients, the duration of treatment is variable and may last three to six months or may be long-term. In the past, the optimal duration of secondary prevention was based on clinical grounds. More recently, the Prolong randomized interventional clinical trial [24] enrolled patients with a first episode of unprovoked VTE, who had been on VKAs for at least 3 months. D-dimer was measured one month after discontinuation of the treatment. Patients with D-dimer below cut-off were left off therapy, and patients with D-dimer above cut-off were randomized to resume or to stop VKAs. All patients were followed up to 1.5 years to record recurrent thrombosis or hemorrhage. The study showed that patients who had D-dimer above cut-off after discontinuation of VKA treatment and did not resume anticoagulation had a significantly high risk of developing recurrent VTE when compared to those with high D-dimer who did resume VKAs [24]. Other studies confirmed that the strategy of measuring D-dimer after the regular course of anticoagulation with VKAs could identify those patients at risk of recurrent VTE who may benefit from extended anticoagulation [25, 26]. Another study concluded that the long-term risk of recurrence in patients with a first unprovoked VTE who have D-dimer below cut-off is not low enough to justify stopping anticoagulants in men, but was low enough in women to choose stopping anticoagulation [27]. Cumulatively, the above studies paved the way for the use of D-dimer as an adjunctive criterion to make decisions on the optimal duration of secondary thromboprophylaxis. There are, however, issues that remain to clarify. For example, it is unclear if D-dimer should be measured during or after discontinuation of anticoagulation. Cut-offs to make decisions are not well established, and it is unclear whether cut-offs used for the VTE exclusion are suitable also to make decisions on the optimal duration of anticoagulation. Finally, whether D-dimer is useful to make decisions on the optimal duration of prophylaxis with VKAs can also be applied to patients on DOACs is not well established. Recently, a protocol similar to the Prolong (Apidulcis) [28] was applied to patients who after unprovoked VTE and treatment with any anticoagulant for at least 12 months underwent D-dimer testing during anticoagulation. If D-dimer was below cut-off, measurements were repeated 15, 30, and 60 days after stopping anticoagulation. Patients with D-dimer below cut-off stopped anticoagulation, and patients with the first D-dimer above cut-off were given low-dose apixaban (2.5 mg, b.i.d.). Recurrent VTE was recorded for 18 months [28]. The majority of patients (91%) at the time of randomization were on DOACs. The study was prematurely terminated because of excess primary outcomes in patients with D-dimer below cut-off who discontinued anticoagulation, indicating that, at variance with the Prolong, Apidulcis was unable to confirm the value of D-dimer to stratify patients for extended anticoagulation [28]. The reasons for the D-dimer failure in the Apidulcis are unknown. It was hypothesized that since the study was carried out during the COVID-19 pandemic, enrolled patients had an increased VTE risk because of limited physical activity imposed by the lockdown [29]. Recently, an alternative explanation was proposed. Perhaps the type of anticoagulation (VKAs vs. DOACs) used in the Prolong and the Apidulcis studies may impact differently on D-dimer levels achieved during anticoagulation [30]. It was in fact shown that patients on DOACs display D-dimer levels that are substantially higher than those achieved in patients on VKAs [30]. Accordingly, it cannot be excluded that the poor value of D-dimer to make decisions on extended anticoagulation in patients on DOACs, as observed in the Apidulcis, is due to the poor sensitivity of this biomarker to detect biochemical signs of hypocoagulability achieved by DOACs. Whatever the reasons, the results so far accumulated indicate that D-dimer is less effective to stratify patients for extended anticoagulation when they are on DOACs. Clinical trials with appropriate design are warranted to resolve definitively this issue. Until then, D-dimer should not be used to make decisions on extended anticoagulation in patients on DOACs.

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STATEMENTS

- We *suggest the use of D-dimer after stopping the regular course of anticoagulation with VKAs* as an adjunct to clinical criteria to make decision on the optimal duration of secondary prevention of VTE when there is an equipoise between risks and benefits of anticoagulation (*first voting round 25% strongly agreed + 41.7% agreed = 66.7%*).

Remarks

The statement was close, but did not reach the predefined 75% acceptance rate. The general feeling was that in the first voting round, the panelists did not consider appropriately that the use of D-dimer was suggested as an adjunct to clinical criteria, not as a standalone criterion and when there is a balance between risks and benefits of anticoagulation. Hence, the statement was reformulated to state unequivocally the intended use of D-dimer.

- *Modified statement*. We *suggest the use of D-dimer after stopping the regular course of anticoagulation with VKAs* as an adjunct to clinical criteria to make decision on the optimal duration of secondary prevention of VTE when there is a balance between risks and benefits of anticoagulation (*second voting round 45.5% strongly agreed + 45.5% agreed = 91%*).
- We *strongly suggest against using D-dimer after stopping the regular course of anticoagulation with DOACs* to make decision on the optimal duration of secondary prevention of VTE (*8.3% strongly agreed + 41.7% agreed = 50%*).

Remarks

This statement was far from the predefined 75% acceptance rate. The general feeling is that the panelists could rely only on the Apidulcis study, whose results and conclusions are probably affected by the COVID-19 pandemic. Hence, the statement was reformulated changing *we strongly suggest* for *we suggest* to state unequivocally that there are currently insufficient data to allow for the safe use of D-dimer in that clinical context.

Modified Statement. We *suggest against using D-dimer after stopping the regular course of anticoagulation with DOACs* to make decision on the optimal duration of secondary prevention of VTE until new data are provided by randomized clinical trials on the validity of D-dimer (*second voting round, 55.6% strongly agreed + 33.3% agreed = 88.9%*).

3.3 | D-Dimer Use in Patients With Suspected Acute Aortic Syndrome

Acute aortic syndrome defines a group of potentially fatal diseases that affect the thoracic aorta (i.e., aortic dissection, intramural aortic hematoma, penetrating aortic ulcer, and aortic rupture) [31]. Acute aortic syndrome represents a clinical emergency with high rates of morbidity and mortality and, thus, a rapid diagnosis is crucial to reduce adverse outcomes. The most common clinical presentations are non-specific and include truncal pain, syncope, neurological deficits, and limb ischemia, while signs such as pulse deficit are more specific but less common [32]. Pre-test probability scores were developed to improve the initial assessment of patients with suspicion of acute aortic syndrome, and the Aortic Dissection Detection Risk Score (ADD-RS) is the most validated [33]. This score includes predisposing conditions, pain type, and physical examination. Patients with a score equal to or lower than 1 are classified as low probability of disease, but the score alone is insufficient to safely rule out acute aortic syndrome. Hence, the combination of the ADD-RS score with D-dimer was proposed [34]. D-dimer levels increase in patients with acute aortic syndrome, and negative D-dimer results in combination with low pretest pre-test probability may safely rule out the disease with an acceptably low false-negative rate [35]. In the ADvISED study, the combination of negative D-dimer and an ADD-RS score of 0 yielded a failure rate of 0.3% (95% CI, 0.1–1.9) and an efficiency of 15.9 (95% CI, 14.3–17.6), while a negative D-dimer and an ADD-RS score of 0 or < 1 yielded a failure rate of 0.3% (95% CI, 0.1–1.1) and an efficiency of 49.9% (95% CI, 47.7–52.2) [35]. A number of metanalyses were carried out to determine the diagnostic accuracy of D-dimer for acute aortic syndrome. In the most recent study, including 18 studies, D-dimer showed a sensitivity of 96.5% with a specificity of 56.2% [36]. The potential for false-negative results in very early or in late presenters, as well as in patients with intramural hematoma and focal dissections, was reported. The possibility of improving the accuracy of D-dimer testing using age-adjusted D-dimer levels was proposed, but additional evidence is probably required before this strategy can be routinely implemented in clinical practice [37].

STATEMENT

- *We suggest the use of D-dimer in combination with the ADD-RS pre-test probability score* to safely rule out acute aortic syndromes in the emergency setting without further imaging tests (*25% strongly agreed + 50% agreed = 75%*).

4 | Use of D-Dimer Measurement Not Yet Substantiated by Evidence

There are situations where the clinical value of D-dimer has been advocated, but not yet substantiated by clinical observations. For the purpose of this guidance, we elaborated a set of seven questions that may represent possible scenarios encountered in clinical practice.

PICO 1. In patients with first incidental finding of unexplained elevation of D-dimer and no known risk factors for VTE, how do you rate appropriateness to prescribe pharmacological prophylaxis of VTE?

D-dimer measurement was proposed in some patients at risk of VTE to drive the choice, intensity, and duration of pharmacological primary prophylaxis. For example, in patients hospitalized because of COVID-19, high D-dimer was associated with poor prognosis, and some studies suggested the need for an intermediate or therapeutic dose of LMWH over standard prophylactic doses [38–42]. In hospitalized medically ill patients, high D-dimer at admission or at the time of discharge was associated with an increased VTE risk after hospitalization

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and was proposed as a possible additional strategy to identify patients who may benefit from an extended duration of thromboprophylaxis [43–46]. Finally, D-dimer was proposed in addition to clinical risk assessment models (i.e., Khorana risk score, IMPEDE VTE score) to better identify ambulatory patients with cancer who may benefit from pharmacologic prophylaxis [47, 48]. Conversely, there is no evidence to suggest a higher risk of subsequent VTE in patients with a first incidentally detected elevation of D-dimer and no additional risk factors for VTE. Likewise, there is no study to support a benefit for prophylactic strategies in these patients.

STATEMENT
- We *strongly suggest against the prescription of pharmacological prophylaxis* in patients with a first incidental finding of unexplained elevation of D-dimer without known risk factors for VTE (*83.3% strongly agreed*).

PICO 2. In patients with the first incidental finding of unexplained elevation of D-dimer, how do you rate the appropriateness of searching by CUS or CT scan for clinically unsuspected venous thromboembolism (VTE)?

The role of elevated D-dimer as a predictor of future incident VTE in the general population was recognized by prospective studies showing a 1.5–4-fold increase in risk in subjects with the highest levels. In these studies, D-dimer was measured at baseline in VTE-free subjects who underwent a prolonged follow-up (8–17 years) [49–53]. However, no evidence is available that in subjects with an unexplained elevation of D-dimer, in the absence of symptoms or signs of VTE, performing CUS or CT scan helps diagnose otherwise unsuspected VTE. On the other hand, this issue is often addressed in the clinical practice, with D-dimer testing misused, as also recently highlighted during and following the COVID-19 pandemic [54]. In the absence of evidence-based management, according to expert opinions, patient assessment should consider known determinants of D-dimer increase (including physical exercise, recent/ongoing pregnancy, obesity) and possible overlooked or currently resolved symptoms [55].

STATEMENT
- We *strongly suggest against performing CUS or CT-scan for clinically unsuspected VTE* in patients with first incidentally detected and unexplained high D-dimer (*41.7% strongly agreed + 58.3% agreed = 100%*).

PICO 3. In patients with the first incidental finding of unexplained elevation of D-dimer, how do you rate the appropriateness of repeating testing after at least two or 3 months?

Several causes other than VTE may raise D-dimer (Figure 2). The relevance of otherwise unexplained high D-dimer in outpatients with no clinical suspicion of VTE is unclear. However, it should be acknowledged that very high D-dimer values (e.g., > 5000 μg/L) rarely are rarely associated with a specific disease [56].

Data on the usefulness of D-dimer repeat testing in cases of unexplained D-dimer elevation is scanty. Once all possible known causes of increased D-dimer have been excluded, if the patient has no recent specific symptoms of VTE and the physical examination is normal, a careful clinical follow-up may be suggested.

STATEMENT
- We *suggest against repeat testing* in patients with unexplained elevation of D-dimer levels and no specific symptoms or clinical signs of VTE (*50% strongly agreed + 25% agreed = 75%*).

PICO 4. In patients with the first incidental finding of unexplained elevation of D-dimer, how do you rate the appropriateness of performing extensive or limited cancer screening?

The close relationship between VTE and cancer was recognized since 1865, when Armand Trousseau [57] first identified unprovoked VTE as a potential initial manifestation of occult cancer. A systematic review and meta-analysis of individual patient data estimated the 12 months prevalence of cancer following unprovoked VTE to be 5.2% [58].

Despite ongoing efforts to develop an effective screening strategy for occult malignancy in patients with unprovoked VTE, uncertainty remains regarding the clinical benefits of cancer screening. While current evidence suggests that screening may enable earlier cancer detection, no high-quality studies have demonstrated an improvement in survival outcomes resulting from early detection.

A limited cancer screening including medical history, physical examination, basic blood work, chest radiography, abdomen US, and age- and sex-specific testing is currently suggested for patients with unprovoked VTE [59]. The main goal of this strategy is to detect cancer at an early and curable stage, which may potentially reduce cancer-related morbidity and mortality.

Whether a more extensive screening, for example with computed tomography of the abdomen or whole-body positron emission tomography, should be systematically implemented in all cases is debated. An extensive screening strategy appears to increase the detection of occult cancer by two-fold compared with a more limited approach, although this does translate into a higher detection rate of early-stage cancer [58]. Extensive screening may be burdensome for patients and healthcare systems because of additional, sometimes invasive, procedures, which expose patients to procedure-related complications and may increase patient distress and costs.

Clinical and laboratory factors, including older age, smoking, prior VTE, male sex, chronic lung disease, anemia, and elevated platelet counts, have been associated with the diagnosis of occult cancer in patients with VTE, and could help target the use of extensive screening tests in high-risk subgroups [58, 60].

These considerations apply even more so to D-dimer. Studies in ambulatory or hospitalized patients with VTE suggested that elevated D-dimer is associated with a higher risk of occult cancer, although results were inconsistent [59–66]. Various D-dimer

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thresholds were used in these studies, with no standardized workup for occult cancer.

In a retrospective cohort of ambulatory and hospitalized patients with D-dimer above 5000 μg/L, cancer was the second most prevalent disease (29%) after VTE (32%) [56]. While these findings may suggest that very high D-dimer could hint at an underlying cancer, the heterogeneity of the study population as well as the poor reporting on patients' clinical presentation, reasons for requesting D-dimer testing, and subsequent workup limits these conclusions.

Therefore, the most important screening components for occult cancer in individuals with incidentally detected elevated D-dimer remain the thorough medical history and physical examination to rule out physiological or pathological causes of increased D-dimer as well as suspicion of underlying cancer. Given the costs and potential complications associated with other investigations, clinicians may consider forgoing additional screening cancer procedures in these individuals when no specific suspicion arises from medical history and clinical evaluation.

STATEMENT

- We *strongly suggest against performing limited or extensive cancer screening* in patients with incidentally detected and unexplained high D-dimer (*66.7% strongly agreed + 33.3% agreed = 100%*).

PICO 5. In patients with first incidental finding of unexplained elevation of D-dimer levels, how do you rate the appropriateness of testing for inherited/acquired venous thrombophilia?

Inherited and/or acquired thrombophilia are conditions associated with hypercoagulability and increased VTE risk. There is, however, no evidence that subjects with thrombophilia and no previous VTE require thromboprophylaxis. On the other hand, thrombophilia may also be incidentally associated with increased D-dimer. Thus, it might be debatable whether the combination of inherited thrombophilia and elevated D-dimer in an otherwise healthy subject increases the VTE risk and benefits from thromboprophylaxis. Indeed, while D-dimer levels predict VTE recurrence in patients with a previous unprovoked VTE with or without an underlying thrombophilia [67], there is no evidence from the literature and practice that the detection of thrombophilia in a subject with an isolated increased D-dimer levels requires anticoagulation. This concept is further supported by the observation that D-dimer testing does not bring useful information on the thrombotic risk in carriers of inherited thrombophilia [68].

STATEMENT

- We *strongly suggest against performing screening for inherited/acquired thrombophilia* in patients with incidentally detected and unexplained high D-dimer (*66.7% strongly agreed + 25% agreed = 91.7%*).

PICO 6. In patients with first incidental finding of unexplained elevation of D-dimer levels, how do you rate it appropriate to test for laboratory risk factors of arterial thrombosis?

Several studies reported that patients with known atherosclerosis (both peripheral artery disease and coronary artery disease) had increased baseline D-dimer levels that were also predictive of increased future adverse cardiovascular events [69–74]. When this relationship was investigated in patients presenting with acute myocardial infarction (AMI), D-dimer was high at the time of hospital admission and also predictive of future adverse cardiovascular events. D-dimer was also elevated in patients with cerebrovascular events (e.g., acute ischemic stroke or transient ischemic attack) in whom it predicted a higher risk of future stroke [69].

The LIPID study randomized patients to placebo or pravastatin after AMI or unstable angina. D-dimer was measured at baseline and at 1 year in 7863 subjects. Median follow-up was 6 years during the trial and 16 years in total. D-dimer predicted long-term risk of arterial events and cardiovascular mortality independently of other risk factors [70].

The REGARDS study recruited 30,239 participants and measured baseline D-dimer in patients with stroke or cardiovascular artery disease and in a cohort of random samples. Each doubling of D-dimer level was associated with an increased risk of stroke [HR 1.15 (95% CI 1.01–1.31)] or cardiovascular artery disease (HR 1.27; 95% CI 1.11–1.45) [71].

Five thousand five hundred and forty-one participants in the Reykjavik Study of nearly 19,000 participants were monitored for a mean duration of 20 years. Measurements were made in 1925 subjects who had a first-ever nonfatal AMI or died of cardiovascular artery disease during follow-up and in 3616 controls. Age and sex-adjusted OR for coronary artery disease per 1 standard deviation higher than baseline D-dimer was 1.01 (95% CI 0.95–1.07). After additional adjustment for conventional cardiovascular risk factors, the corresponding OR for D-dimer was 1.06 (95% CI 1.00–1.13). When combined with the results from previous prospective studies in a random-effects meta-analysis, the overall adjusted OR was 1.23 (95% CI 1.16–1.32) [72].

In 5923 patients with acute coronary syndrome undergoing percutaneous coronary intervention after multivariable adjustment, D-dimer significantly predicted in-hospital mortality [73].

The Glasgow MONICA study recruited random population samples in four different periods. The associations of plasma levels of 15 hemostatic variables with the risk of cardiovascular disease and of total mortality in two random population samples were significant for D-dimer, which was also strongly associated with total mortality after full risk factor adjustment [74].

Finally, the association between atherothrombotic biomarkers and traditional atherosclerotic cardiovascular diseases, with and without adjustment for traditional risk factors, was evaluated in 5789 participants of the Multi-Ethnic Study of Atherosclerosis study over a median follow-up of 14.7 years. The study identified two new variables: a thrombotic factor, defined by fibrinogen, plasmin–antiplasmin complex, factor VIII, D-dimer, and lipoprotein (a), and a fibrinolytic factor, defined by plasminogen and oxidized phospholipids on plasminogen. In a model including both factors, the thrombotic factor was associated with the higher

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risk of atherosclerotic cardiovascular disease [HR 1.57, 95% CI 1.45–1.70], while the fibrinolytic factor was associated with the lower risk of atherosclerotic cardiovascular disease (HR 0.76, 95% 0.70–0.82) [75]. Therefore, based on the available evidence, high D-dimer was consistently associated with a higher risk of atherosclerotic cardiovascular disease. No relevant data indicate that the elevation of D-dimer is associated with other hemostatic variables, such as fibrinogen, homocysteine, or antiphospholipid antibodies. Furthermore, although D-dimer and other hemostatic biomarkers may be important in understanding the pathophysiology of cardiovascular diseases, their value for clinical risk prediction has not yet been investigated by ad-hoc studies, and their usefulness in individual patients is uncertain. For example, it is unclear whether the link between laboratory risk factors and arterial events is causally related. Finally, therapeutic interventions justifying D-dimer measurement are not available.

STATEMENTS

- We *strongly suggest against testing other biomarkers*, such as fibrinogen, FVII, homocysteine, high-sensitivity C-reactive protein, lupus anticoagulant, anticardiolipin and anti-β2-Glycoprotein-I antibodies in patients with first incidental finding of unexplained elevation of D-dimer (*66.7% strongly agreed + 33.3% agreed = 100%*).
- We *strongly suggest against testing D-dimer* in patients with or without a previous arterial event in order to establish the risk of future arterial events (*100% strongly agreed*).
- We *suggest an accurate evaluation and possibly improvement management of traditional cardiovascular risk factors* in patients with incidental elevated D-dimer levels (*66.7% strongly agreed + 25% agreed = 91.7%*).

PICO 7. In patients with moderate-to-high 10 years cardiovascular risk according to validated risk assessment models and first incidental finding of unexplained elevation of D-dimer levels, how do you rate appropriate to perform instrumental tests searching for atherosclerotic disease?

Available data show that D-dimer levels are associated with the presence of peripheral artery disease measured by ankle-brachial index in population-based studies.

In 1291 African-Americans (mean age, 63.4 ± 9.3 years) and 1152 non-Hispanic whites (mean age, 58.5 ± 10.1 years) belonging to hypertensive sibships, D-dimer, in addition to 46 other markers, were measured. D-dimer was significantly associated with ankle-brachial index [76].

In the more than 13,000 participants from the Atherosclerosis Risk in Communities (ARIC) study, plasma levels of D-dimer, in addition to other hemostatic and inflammatory markers, were significantly associated with the presence of peripheral artery disease documented by ankle-brachial index [77].

In the Edinburgh Artery Study, a population cohort study of 1592 men and women aged 55–74 years, D-dimer was associated with peripheral atherosclerotic progression, measured by ankle-brachial index, independently from cardiovascular disease risk factors [78].

No association was documented between D-dimer and subclinical atherosclerosis (carotid artery stiffness) in 1096 men without, and 303 with cardiovascular disease, aged 71–92 years, from the British Regional Heart Study [79].

In the San Diego Population Study of 1103 men and women, D-dimer was consistently associated with markers of peripheral artery disease, including higher common femoral intima-media thickness and lower ankle-brachial index [80].

Finally, a systematic analysis aimed to determine the association between family history of coronary heart disease and markers of subclinical cardiovascular disease showed consistent evidence of a positive family history as an independent predictor of subclinical disease as assessed by several markers, such as coronary artery calcium, carotid intima-media thickness, inflammatory markers, and flow-mediated dilatation [81]. This notwithstanding, current literature does not answer the question on the utility of searching for subclinical atherosclerosis in individual patients presenting with moderate-to-high 10 years cardiovascular risk, according to validated risk assessment models and the first incidental finding of unexplained elevation of D-Dimer.

No data is available linking D-dimer with other markers of subclinical atherosclerosis.

STATEMENTS

- We *strongly suggest against performing instrumental testing aimed at searching for atherosclerotic disease* in patients with moderate-to-high 10 years cardiovascular risk according to validated risk assessment models and first incidental finding of unexplained elevation of D-Dimer levels (*50% strongly agreed + 41.7% agreed = 91.7%*).
- We *suggest searching for peripheral artery disease by the measurement of ankle-brachial index* in patients with moderate-to-high 10 years cardiovascular risk according to validated risk assessment models and first incidental finding of unexplained elevated D-dimer (*25% strongly agreed + 50% agreed = 75%*).

5 | Concluding Remarks

D-dimer can be considered as one of the most powerful tests that emerged over the last three decades in laboratory medicine. The evidence-based use of D-dimer measurement in some conditions has been consolidated. Currently, however, there is an entrenched, unjustified prescription of D-dimer testing to screen otherwise healthy subjects. With this as background, SISET appointed a panel of experts to prepare a guidance on the use of D-dimer testing based on the literature and expert opinion. The main conclusions of the panel can be summarized as follows.

- D-dimer can be reliably used in combination with pre-test clinical probability to exclude VTE in outpatients with clinical suspicion.

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- It may help establish the optimal duration of secondary prevention of VTE with VKAs, although the panel felt that this area needs further investigation when dealing with DOACs.
- D-dimer in combination with the ADD-RS pre-test probability score can be used to safely rule out acute aortic syndromes in the emergency setting without further imaging tests.

In all the other conditions, testing for D-dimer is discouraged.

Author Contributions

All content was prepared by the authors. Armando Tripodi wrote the final manuscript, which was approved by the authors.

Acknowledgments

Armando Tripodi, while preparing this manuscript, was supported in part by the Italian Ministry of Health, Bando Ricerca Corrente 2024.

Ethics Statement

Approval from ethics committee is not required. The manuscript is a review of the current literature on the topic and does not report results of original experimental investigations performed by the authors. It has been approved by the institutional review boards of the authors.

Conflicts of Interest

Armando Tripodi received honoraria for lectures from Biomarin, Stago, Werfen. Walter Ageno received honoraria for lectures from BMS, Pfizer, Astra Zeneca. Luca Barcella received grants from participation to congresses from BioVIIIx, Fresenius Kabix, CLS Vifor, Mallinckrodt Pharmaceutical. Antonio Coppola received consulting fees from Bayer, Roche, Sobi and honoraria for lectures from Biomarin, CLS Behring. Marcello Di Nisio received research grants from Leo Pharma, consulting fees from Leo Pharma and Janssen and honoraria for lectures from Daiichi Sankyo, Mylan and Werfen. Rossella Marcucci received research grants from Novartis and honoraria for lectures from Menarini, Sanofi, Amgen, Viatris, Novartis, Amarin and participation to Novo Nordisk, Amarin Advisory Boards. Valerio De Stefano received research grants from, Alexion, Novartis; consulting fees from AOP-Health, Novartis; honoraria for lectures from AbbVIE, Alexion, Amgen, AOP-Health, Bristols Myers Squibb, Grifols, GSK, Novartis, Novo Nordisk, Sanofi, Takeda. Massimo Franchini, Monia Marchetti, Marco Marietta, Pasquale Pignatelli and Domenico Prisco declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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